Economics 5040: Dynamic Stochastic Modeling
Problem Set 1 — Intertemporal Choice, Production, and Equilibrium

Instructions

Submit one Jupyter notebook containing all derivations, Python code, figures, numerical answers, and economic explanations.

You are encouraged to use Python extensively. You may also consult textbooks, lecture notes, classmates, and AI tools. However, your notebook must make clear that **you understand every result you report**. A numerical answer without an economic explanation will receive little credit.

For every optimization problem:

- state the objective and constraints;
- derive the relevant first-order condition analytically when possible;
- solve the problem numerically in Python;
- check that the numerical solution satisfies the theoretical optimality condition;
- explain the economics of the result.

Use clear figures with labeled axes and legends.

Problem 1. Utility, Rankings, and Intertemporal Preferences

Consider the two-period utility function

$$U(c_0, c_1) = \log c_0 + \beta \log c_1, \qquad \beta = 0.95.$$

(a)

Write a Python function that evaluates $U(c_0, c_1)$.

Evaluate the utility of the following consumption plans:

$$A = (4, 4), \qquad B = (3, 6), \qquad C = (5, 3).$$

Rank the three plans.

(b)

Now define

$$V(c_0, c_1) = \exp\left(U(c_0, c_1)\right).$$

Use Python to rank the same three consumption plans according to V.

Explain why the ranking must be identical to the ranking generated by U.

(c)

Create a grid of at least 10,000 positive consumption bundles in the (c_0, c_1) plane.

Compute U and V at every point and use Python to verify numerically that whenever

$$U(x) > U(y),$$

the transformed utility function also gives

$$V(x) > V(y).$$

Explain what this exercise illustrates about utility functions and preferences.

(d)

Suppose instead that preferences over three dated goods are represented by

$$U(c_0, c_1, c_2) = \log c_0 + \beta \log c_1 + \beta^2 \log c_2.$$

Derive the marginal rate of substitution between c_1 and c_2.

Show explicitly that c_0 does not appear in it.

In a short paragraph, explain why this illustrates both the usefulness and the restrictiveness of time-separable preferences.

Problem 2. A Two-Period Exchange Economy

A consumer has the endowment

$$(e_0, e_1) = (4, 6)$$

and preferences

$$U(c_0, c_1) = \log c_0 + \beta \log c_1, \qquad \beta = 0.96.$$

The gross real interest rate is R.

(a)

Write the consumer's present-value budget constraint.

Show that its slope in (c_0, c_1)-space is

$$-R.$$

Explain carefully why R, rather than $1/R$, is the relative price of current consumption in units of future consumption.

(b)

Let

$$R = 1.10.$$

Solve the consumer's problem analytically.

Then solve it independently using a numerical optimizer in Python.

Report

$$c_0^*, \qquad c_1^*,$$

and saving

$$s = e_0 - c_0^*.$$

Check numerically that

$$\frac{u'(c_0^*)}{\beta u'(c_1^*)} = R.$$

(c)

Produce a figure showing:

- the endowment;
- the budget line;
- the budget set;
- the optimal consumption bundle;
- the indifference curve through the optimum.

Your graph should make clear whether the consumer is a borrower or a saver.

(d)

Solve the problem for 100 values of R uniformly spaced between

$$0.8 \leq R \leq 1.5.$$

Plot $c_0^*(R)$, $c_1^*(R)$, and $s(R)$.

Do not simply describe the plots. Explain why the consumer's response to R reflects both an income effect and a substitution effect.

(e)

Find numerically the value $R = \widehat{R}$ for which the consumer chooses exactly the endowment:

$$(c_0^*, c_1^*) = (e_0, e_1).$$

Then derive $\widehat{R}$ analytically.

Explain its economic meaning.

Problem 3. Intertemporal Production and the Planner

Consider an economy with initial output

$$y_0 = 10.$$

Current output can either be consumed or invested:

$$c_0 + k_0 = y_0.$$

Investment produces future output according to

$$c_1 = f(k_0) = Ak_0^{\alpha},$$

where

$$A = 2.5, \qquad \alpha = 0.4.$$

The representative consumer has preferences

$$U(c_0, c_1) = \log c_0 + \beta \log c_1, \qquad \beta = 0.95.$$

(a)

Write the planner's problem explicitly as

$$\max_{c_0, c_1, k_0} U(c_0, c_1)$$

subject to the economy's feasibility constraints.

Then reduce the problem to an optimization problem with only k_0 as the choice variable.

(b)

Derive the planner's first-order condition.

Show that the efficient allocation satisfies

$$MRS = MRT.$$

Write both sides explicitly for the functional forms in this problem.

(c)

Solve analytically for

$$k_0^*, \qquad c_0^*, \qquad c_1^*.$$

Then solve the same problem numerically in Python.

The two answers should agree to at least six decimal places.

(d)

Plot the intertemporal production possibility frontier

$$c_1 = f(y_0 - c_0).$$

On the same graph show:

- the planner's optimal allocation;
- the indifference curve through that allocation;
- a supporting tangent line.

Calculate the slope of all three objects at the optimum where appropriate.

(e)

Change β over the range

$$0.50 \leq \beta \leq 1.00.$$

For at least 100 values of β, calculate the optimal level of investment.

Plot

$$k_0^*(\beta).$$

Explain the result economically.

Then answer a harder question: is the relationship necessarily monotone for every concave utility function and every concave production function? Explain your reasoning.

Problem 4. Decentralizing the Planner's Allocation

Continue to use the economy in Problem 3.

Suppose now that production is carried out by a competitive firm and consumption is chosen by a competitive consumer.

Let R denote the gross real interest rate.

(a)

The firm solves

$$\max_{k_0 \geq 0} \left[\frac{f(k_0)}{R} - k_0 \right].$$

Derive the firm's first-order condition.

(b)

Using the planner's optimal k_0^*, calculate the gross interest rate

$$R^* = f'(k_0^*).$$

Report the corresponding net real interest rate

$$r^* = R^* - 1.$$

(c)

Take R^* as given and solve the consumer's intertemporal optimization problem.

You will need to specify the consumer's wealth correctly, including any income associated with ownership of the firm.

Show numerically that the consumer chooses the same allocation as the planner.

(d)

Verify all three equalities numerically:

$$\frac{u'(c_0^*)}{\beta u'(c_1^*)} = R^* = f'(k_0^*).$$

Report the numerical error in each equality.

(e)

Explain in your own words what has happened.

Your explanation must distinguish clearly among:

1. the planner's problem;
2. the consumer's optimization problem;
3. the firm's optimization problem;
4. market clearing.

Why is the coincidence of the planner's allocation and the competitive allocation economically interesting?

Problem 5. A Computational Experiment with Disequilibrium

The equality

$$MRS = R = MRT$$

holds in competitive equilibrium. This problem asks what happens when R is wrong.

Continue to use the economy from Problems 3 and 4.

(a)

Construct a Python function that takes an arbitrary value of R and returns:

- the firm's desired investment $k^d(R)$;
- the consumer's desired saving $s(R)$.

Define an excess-demand or market-clearing function

$$Z(R) = k^d(R) - s(R).$$

Be explicit about your sign convention.

(b)

Evaluate $Z(R)$ on a fine grid of economically sensible interest rates.

Plot

$$Z(R)$$

against R.

Mark the horizontal line $Z = 0$.

(c)

Use a root-finding algorithm such as bisection or Brent's method to solve

$$Z(R) = 0.$$

Do not simply call the routine. Explain why a root of this equation is a competitive equilibrium.

(d)

Compare the equilibrium interest rate found by market clearing with the interest rate obtained in Problem 4 from

$$R^* = f'(k_0^*).$$

They should coincide numerically.

Explain why this agreement is not an accident.

(e)

Choose one value

$$R_L < R^*$$

and another

$$R_H > R^*.$$

For each one, report household saving and firm investment.

Explain the direction of the market imbalance.

Then explain how a change in the real interest rate would tend to eliminate that imbalance.

Problem 6. The First Welfare Theorem — What the Computer Does Not Prove

The computations above suggest that the competitive equilibrium allocation is efficient.

(a)

Use Python to generate 100,000 random feasible allocations satisfying

$$c_0 + k_0 \leq y_0, \qquad c_1 \leq f(k_0).$$

Evaluate utility at every allocation.

Compare the best randomly generated allocation with the planner's optimum.

(b)

Does your numerical experiment prove that the planner's solution is globally optimal?

Explain why or why not.

(c)

Does the fact that the competitive allocation coincides numerically with the planner's solution constitute a proof of the First Welfare Theorem?

Explain why or why not.

(d)

State the First Welfare Theorem in words.

Then explain what Pareto efficiency means in this one-consumer economy.

(e)

The First Welfare Theorem is frequently summarized by saying that "markets are efficient."

Explain why this slogan is too loose.

Your answer should mention at least two assumptions or qualifications that might matter when applying the theorem to an actual economy.

Problem 7. Synthesis

Write approximately 500 words answering the following question:

> In this two-period economy, what economic role is played by the gross real interest rate?

Your answer must explain the relationship among

preferences, technology, prices, optimization, market clearing,

and

Pareto efficiency.

You should explain—not merely quote—the significance of

$$\boxed{MRS = R = MRT.}$$

You may use numerical results or figures from the earlier problems to support your argument.

A Final Requirement

At the end of your notebook, include a short section entitled

What surprised me?

Identify one result from this assignment that you did not find obvious before doing the computations. Explain why it surprised you and what you now understand differently.

There is no single correct answer to this question, but a generic response will receive no credit.